

A Design space

personality composition as the independent variable

trait level



trait heterogeneity



response surface



Llama-3.1-8B

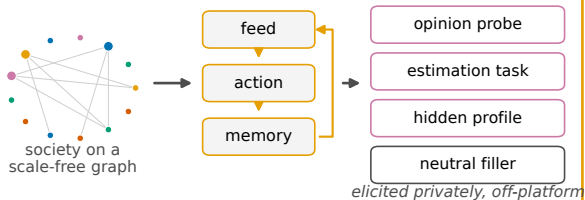
primary

+5 models

3B to 32B

B Simulation pipeline

N agents, T rounds, recommender-driven feed



991 runs $\cdot$ 6 models $\cdot$ 6 topics $\cdot$ 8 seeds

induction gate per model $\cdot$ filler topic $\cdot$ realised traits
probe and recommender ablations $\cdot$ perturbation audit

C Two outcome families, one run

POLARIZATION

dispersion

variance
extremity
bimodality

segregation

cross-cutting
echo closure
assortativity

COLLECTIVE INTELLIGENCE

estimation

collective error
diversity term
median error

decision

hidden profile
pre/post shift

measured on the same societies, in the same runs

both measured in every run

D What composition does

	O $\uparrow$	C $\uparrow$	E $\uparrow$	A $\uparrow$	N $\uparrow$
Opinion variance	-0.31			-0.47	+0.28
Extremity	+0.15	+0.22		+0.20	+0.23
Cross-cutting		-0.06		-0.12	
Collective accuracy					

shaded cells survive Holm correction; intensity $\propto |d_z|$